

Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN may over-ionize helium, constraining super-Eddington accretion to a small fraction.

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Background

JWST found many faint AGN at $z > 4$, called Little Blue Dots, possibly super-Eddington. Their ionizing output could affect helium reionization timing.

How super-Eddington accretion in Little Blue Dots impacts helium reionization is unknown.

What this paper is selling

The paper shows that if most Little Blue Dots are super-Eddington, helium reionizes too early. Observations limit this fraction to under 10%.

Key results to remember

- Super-Eddington LBDs would reionize helium before $z \approx 3$, conflicting with data.
- Only $\leq 10\%$ of LBDs can be super-Eddington under fiducial assumptions.
- If super-Eddington, escape fractions must be $\leq 1.5\%$ or abundance lower.

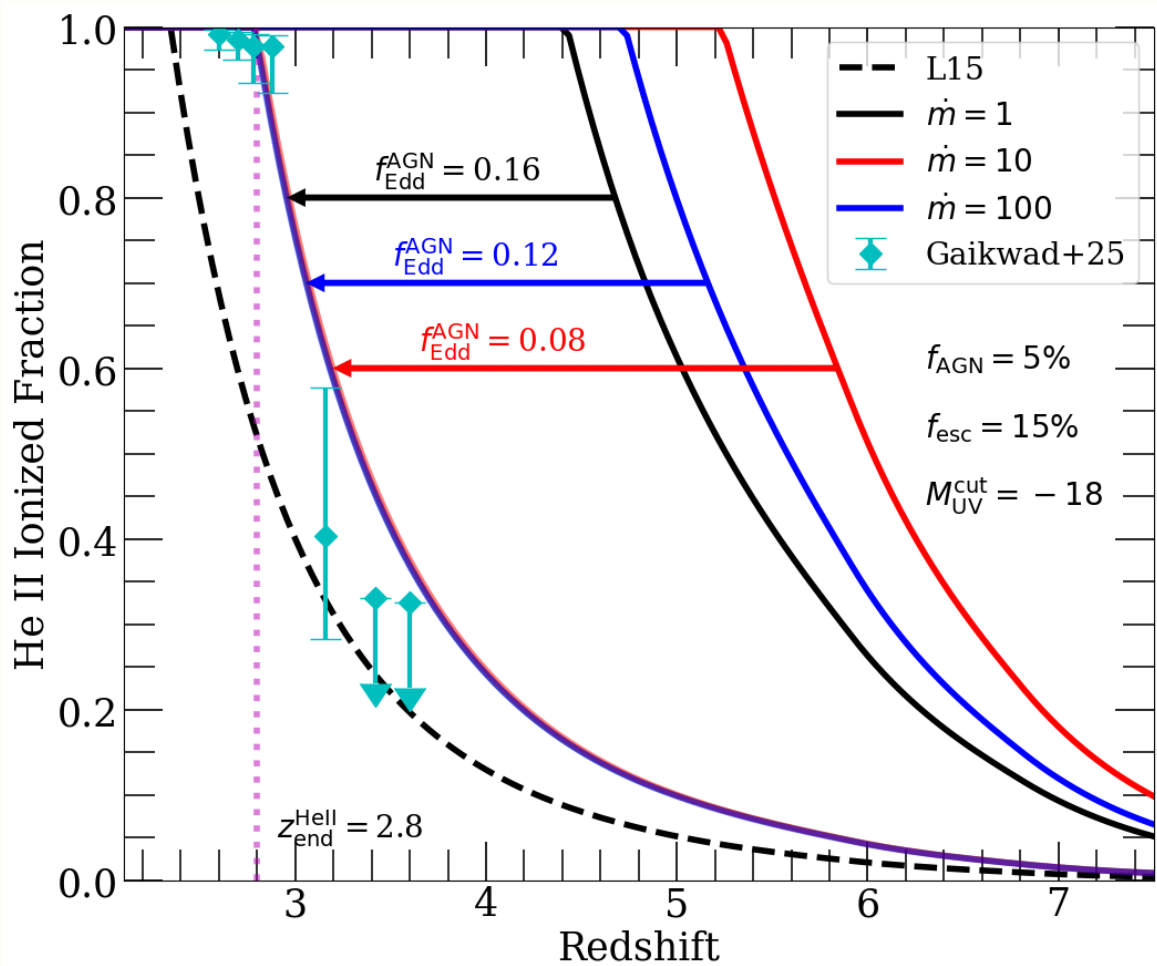


Figure 3 | He\,ii reionization histories for different SED models assuming $f_{\rm AGN} = 0.05$, $f_{\rm esc} = 0.15$, and $M_{\rm UV}^{\rm cut} = -18$.

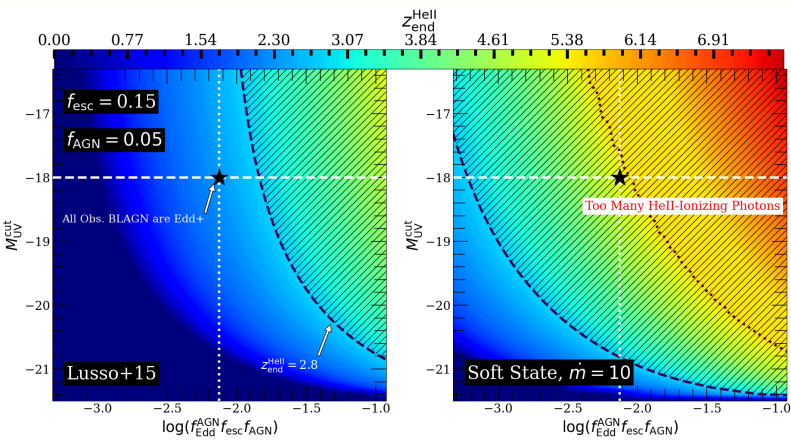


Figure 4 | AGN parameter space excluded by constraints on He\,ii reionization.